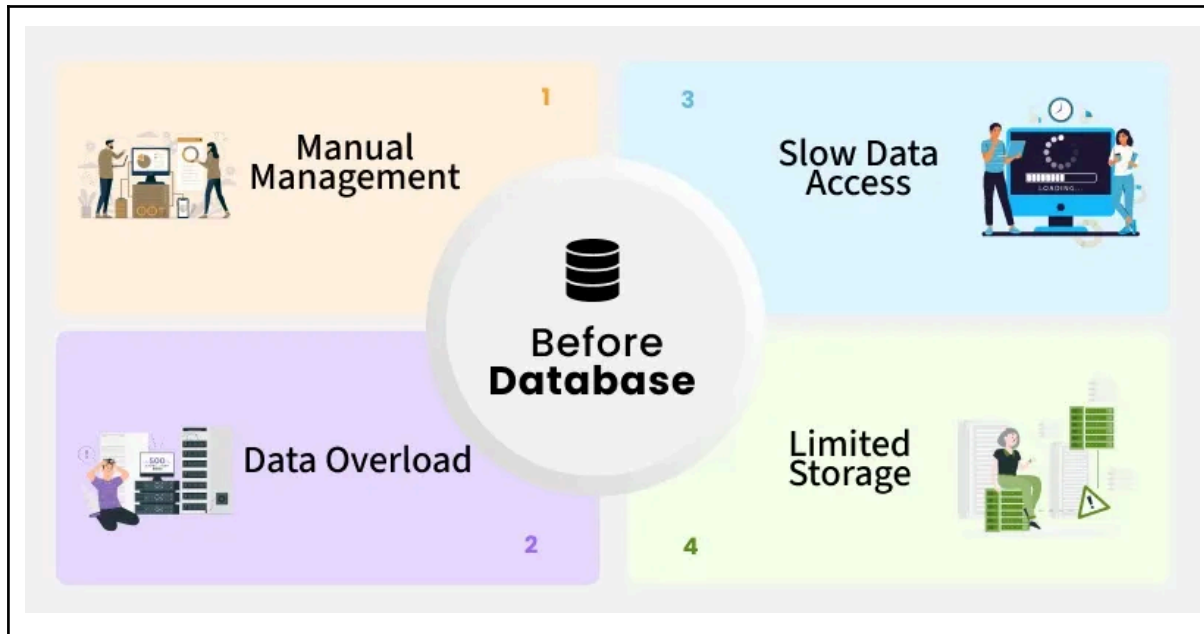


Databases

Databases are systems that allow users to store and organize data. They are technically useful when dealing with large amounts of data.



Who uses databases?

Databases have a wide variety of users:

- Analysts:
 - Marketing
 - Business
 - Sales
- Technical:
 - Data Engineers
 - Data Scientist
 - Software Engineers
 - Web Developers

Basically anyone needing to deal with data

Spreadsheets (Excel) Vs Databases

Most users have some familiarity with spreadsheet software (such as Excel). The use cases of spreadsheets vary from use cases of databases

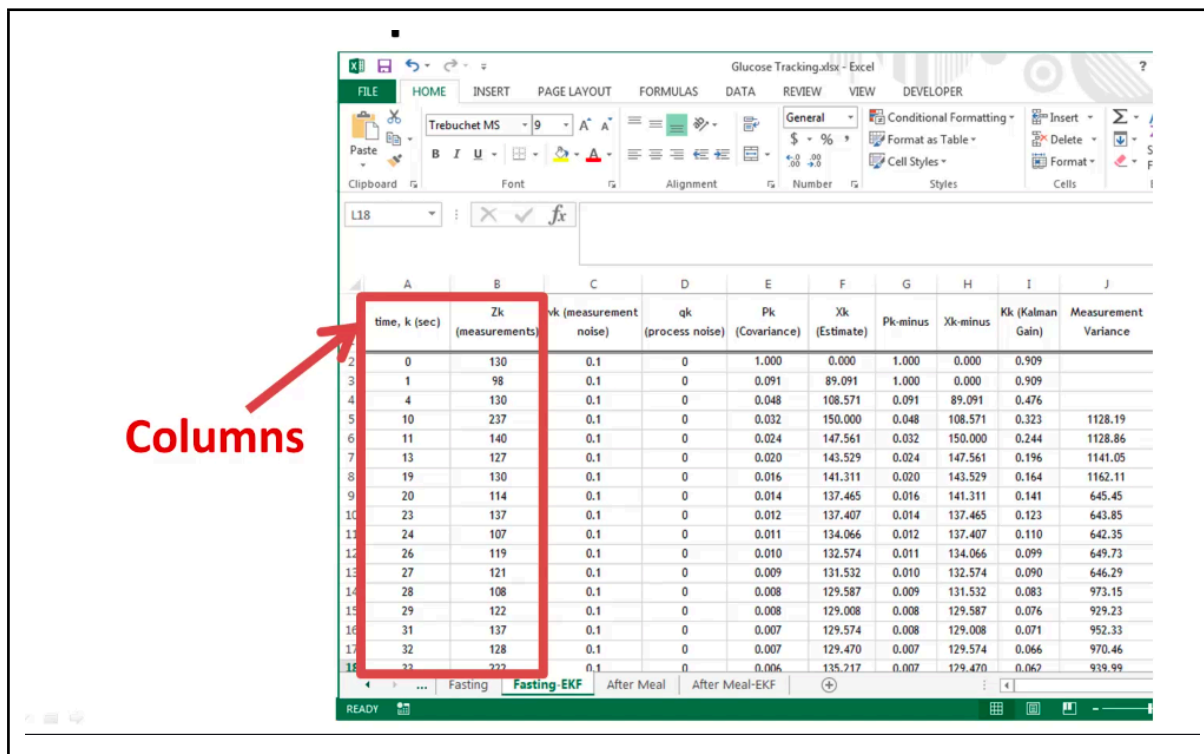
Spreadsheets	Databases
Really good for one time analysis	Data integrity
Quickly need to chart something out	Can handle massive amounts of data
Reasonable dataset size	Can handle massive amounts of data
Ability for untrained people to work with data	Quickly combine different datasets
	Automate steps for re-use
	Can support data for websites and applications

Most users have some familiarity with spreadsheet software (such as excel), so they can easily leverage that knowledge from spreadsheets to understand databases.

Excel Tabs -> SQL Table

The screenshot shows an Excel spreadsheet titled 'Glucose Tracking.xlsx'. The main sheet contains a table with the following columns: time, k (sec), Zk (measurements), vk (measurement noise), qk (process noise), Pk (Covariance), Xk (Estimate), Pk-minus, Xk-minus, Kk (Kalman Gain), and Measurement Variance. The data rows show glucose measurements over time. At the bottom, there are four tabs: 'Fasting', 'Fasting-EKF', 'After Meal', and 'After Meal-EKF'. A red arrow points from the text 'Tabs = Tables' to the 'Fasting' tab, illustrating that each tab in Excel represents a table in a database.

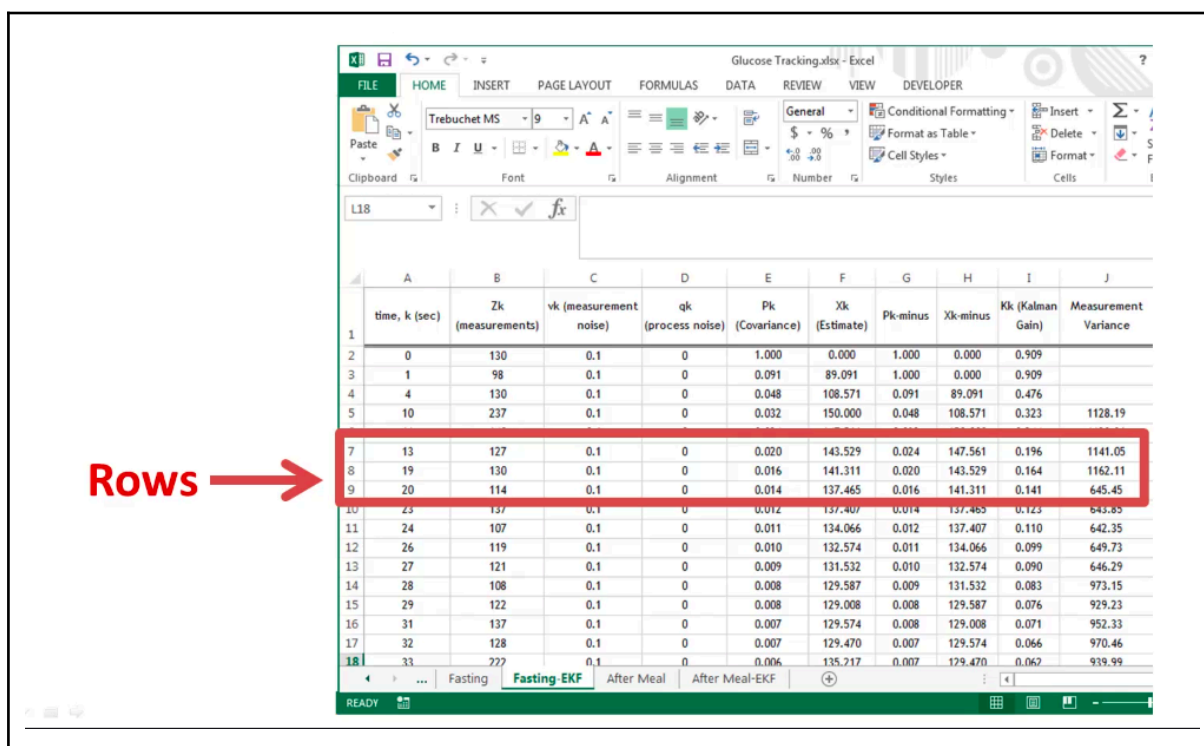
Excel Columns -> SQL Columns



Columns

	A	B	C	D	E	F	G	H	I	J
	time, k (sec)	Zk (measurements)	vk (measurement noise)	qk (process noise)	Pk (Covariance)	Xk (Estimate)	Pk-minus	Xk-minus	Kk (Kalman Gain)	Measurement Variance
2	0	130	0.1	0	1.000	0.000	1.000	0.000	0.909	
3	1	98	0.1	0	0.091	89.091	1.000	0.000	0.909	
4	4	130	0.1	0	0.048	108.571	0.091	89.091	0.476	
5	10	237	0.1	0	0.032	150.000	0.048	108.571	0.323	1128.19
6	11	140	0.1	0	0.024	147.561	0.032	150.000	0.244	1128.86
7	13	127	0.1	0	0.020	143.529	0.024	147.561	0.196	1141.05
8	19	130	0.1	0	0.016	141.311	0.020	143.529	0.164	1162.11
9	20	114	0.1	0	0.014	137.465	0.016	141.311	0.141	645.45
10	23	137	0.1	0	0.012	137.407	0.014	137.465	0.123	643.85
11	24	107	0.1	0	0.011	134.066	0.012	137.407	0.110	642.35
12	26	119	0.1	0	0.010	132.574	0.011	134.066	0.099	649.73
13	27	121	0.1	0	0.009	131.532	0.010	132.574	0.090	646.29
14	28	108	0.1	0	0.008	129.587	0.009	131.532	0.083	973.15
15	29	122	0.1	0	0.008	129.008	0.008	129.587	0.076	929.23
16	31	137	0.1	0	0.007	129.574	0.008	129.008	0.071	952.33
17	32	128	0.1	0	0.007	129.470	0.007	129.574	0.066	970.46
18	33	227	0.1	0	0.006	135.217	0.007	129.470	0.062	939.99

Excel Rows -> SQL Records



Rows

	A	B	C	D	E	F	G	H	I	J
	time, k (sec)	Zk (measurements)	vk (measurement noise)	qk (process noise)	Pk (Covariance)	Xk (Estimate)	Pk-minus	Xk-minus	Kk (Kalman Gain)	Measurement Variance
1										
2	0	130	0.1	0	1.000	0.000	1.000	0.000	0.909	
3	1	98	0.1	0	0.091	89.091	1.000	0.000	0.909	
4	4	130	0.1	0	0.048	108.571	0.091	89.091	0.476	
5	10	237	0.1	0	0.032	150.000	0.048	108.571	0.323	1128.19
6	11	140	0.1	0	0.024	147.561	0.032	150.000	0.244	1128.86
7	13	127	0.1	0	0.020	143.529	0.024	147.561	0.196	1141.05
8	19	130	0.1	0	0.016	141.311	0.020	143.529	0.164	1162.11
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18	33	227	0.1	0	0.006	135.217	0.007	129.470	0.062	939.99

Database Platform Options

Main Ones (Not all of them) :

PostgreSQL		Free (Open Source) Widely used on internet Multi platform
MySQL MariaSQL	 	Free (Open Source) Widely used on internet Multi platform.
MS SQL Server Express		Free, but with some limitations Compatible with SQL Server Windows only
Microsoft Access		Cost (-) Not easy to use just SQL (-)
SQLite		Free (Open Source) Mainly command line (-)